

Reflective Question (Industry relevant)

Automated Visual Defect Detection with CNNs

Industry Context: Manufacturing Quality Control

1. Business Scenario

You are a machine learning engineer at a manufacturing company that produces metal/cast components. Currently, defect inspection is done manually by line workers, which is slow, inconsistent, and expensive at scale. Your task is to build a CNN-based visual inspection system that can be deployed on the production line to automatically flag defective units in real time.

Business goal: Minimize false negatives (missed defects escaping to customers) while keeping false positives low enough that the line doesn't get flooded with unnecessary manual re-checks.

2. Dataset

Use the **Casting Product Image Data** dataset from Kaggle, a real industrial dataset containing submersible pump impeller casting images labeled as defective or ok.

Dataset link: <https://www.kaggle.com/datasets/ravirajsinh45/real-life-industrial-dataset-of-casting-product>

3. Project Tasks

Phase 1 — Baseline CNN Classifier

- Build a CNN from scratch to classify images as **defective vs. non-defective**.
- Establish a baseline: accuracy, precision, recall, F1, and confusion matrix.
- **Share Key lessons and reflection about the learning journey**

Phase 2 — Improve on the Baseline

- Improve on the baseline model using any technique(s) you think are justified — the choice of approach is part of the exercise (e.g. transfer learning, a deeper/wider architecture, better augmentation, addressing class imbalance, hyperparameter tuning). Do not limit yourself to the given examples.
- Justify why you chose this approach over the alternatives you considered.
- Compare your improved model with the one built in Phase 1, using the same test set and metrics.
- **Share key lessons and reflection about the approach you chose and how it performed against the baseline.**

4. Deliverables

Deliverable	Description
Executed Jupyter notebook	Data preparation, CNN model development, training, evaluation, results, graphs, comments, and observations, etc
Model comparison report and reflection	Comparison and discussion of the developed models, their performance, and reflection on the modelling process, learning, and outcomes.

5. Important Instructions:

Each learner is expected to develop a unique and personalised reflection based on their own experiments, experiences, and understanding of the Deep Learning with Artificial Neural Networks module.

Please take note of the following points:

- Your submission must be original.
- Reflections should not be identical or highly similar to those of any classmates.
- Any work that is copied, shared, or jointly prepared will receive zero marks.
- The reflection must demonstrate your own interpretation, insights, and practical understanding rather than generic or reproduced content.
- If any external resources are consulted, they should be appropriately acknowledged.
- Limited use of digital tools for proofreading or basic support may be permitted, but the ideas, structure, and execution must remain your own and reflect your individual thinking.
- Maintain academic honesty and integrity throughout your work.

Submission Requirements

- You must complete this reflection before the assessment day.
- You are required to upload the following as your submission:
 1. The Jupyter Notebook containing your implementation, meaningful comments, outputs, and observations.
 2. A model comparison report presenting and discussing the performance of the models developed along with the reflection.